Balanced gate shipped end-to-end; the document bars are not reachable in this architecture

Results

Three things were run: the unfinished balanced-gate cascade, a feasibility probe on the document-level bars, and an active-learning label-budget curve. One is a free improvement. The other two say to stop spending on this architecture.

1. The balanced gate, carried through a full cascade

Gate refit with source-balanced weights and `alpha = 1e-2`; heads, tag-threshold rule, read profile and evaluator all unchanged from arm B. Both operating points re-derived. Scored on the eight sealed corpora by the same fixed evaluator.

metric	arm B	balanced gate	Δ
document recall	0.7532	**0.7975**	**+0.0442**
document precision	0.8893	**0.8956**	+0.0062
document specificity	0.8832	0.8792	–0.0040
macro F2	0.6497	**0.6614**	+0.0116
macro recall	0.6482	**0.6938**	**+0.0456**
macro precision	0.6154	**0.6182**	+0.0027
worst priority-tag recall	0.6524	**0.7221**	**+0.0697**
micro F1	0.7313	0.7255	–0.0058
priority macro F0.5	0.7467	0.7434	–0.0033
prediction rate	0.7836	0.8192	+0.0357

Priority tag×corpus gates cleared, of 55 measurable:

	≥ **0.90**	≥ **0.75**
arm B	25	43
balanced gate	**29**	**45**

Better on almost every axis at no cost. Same architecture, same latency (3.916 ms p95), same 57 enabled tags. The two regressions — micro F1 –0.006, priority F0.5 –0.003 — are noise beside a +0.070 gain on the worst priority tag. This is the largest single measured improvement in the project and it is free.

It still does not ship: 29 of 55 gates is short of 55.

2. Feasibility — the bars are not reachable

The precision-view policy wants document precision ≥ 0.90, specificity ≥ 0.85 and recall ≥ 0.85 simultaneously, conclusively, on real-world documents. Walking the measured frontier on the sealed real corpora (5,397 positives / 5,152 negatives):

gate	AUC	best recall @ P≥0.90, sp≥0.85	best precision @ R≥0.85
arm B	0.8453	0.5370	0.7008
balanced + regularised	0.8804	**0.6231**	**0.7664**
bar		**0.85**	**0.90**

No threshold on either gate satisfies all three. The shortfall on recall is **0.227** even for the better gate.

3. Active learning — the curve is flat

The obvious response to (2) is "get more real data from the hard distribution". Measured directly, at increasing label budgets, on a fixed held-out 30% of the sealed real documents:

target-distribution documents added	AUC	best recall @ bars
0	0.8832	0.6279
500	0.8763	0.6358
1,500	0.8731	0.6072
3,000	0.8801	0.6376
5,000	0.8804	0.6255
7,370 *(all available)*	0.8778	0.6157

7,370 labelled documents from exactly the distribution being tested move nothing. Flat within noise, slightly down at the end.

Cross-validated *within* the target distribution — training and testing on the sealed real documents only — the ceiling is **AUC 0.869**, below what the balanced gate already achieves from the training distribution.

TL;DR

- **Adopt the balanced gate.** Document recall +0.044, macro recall +0.046, worst priority-tag recall +0.070, gates cleared 25 → 29 of 55, precision and specificity held, latency unchanged. Free.
- **The document bars are not reachable in this architecture.** Best achievable is recall **0.62** holding precision and specificity (bar 0.85), or precision **0.77** holding recall (bar 0.90).
- **Active learning will not close it.** 7,370 labelled target-distribution documents moved AUC 0.883 → 0.878. The labelling budget would buy nothing.

- **The constraint is the representation, not the data.** Hashed unigram/bigram/shape features over 12,000 characters plateau at AUC ≈ 0.88 on real-world documents. Synthetic data was refuted yesterday; real data is refuted now.
- **Verdict:** `target_infeasible` for the current architecture, with the reachable numbers stated above.

Date	2026-08-26
Author	Ryan Lence
Project	`projects/pii-head-to-head-v1`
Run ID	H1 (follow-up)
Scope	document gate and its cascade; heads unchanged
Outcome	one improvement adopted; target declared infeasible for this architecture

Artifacts

Path	What
`models/cascade_balanced/`	the adopted gate, full cascade
`evaluations/arm_cascade_balanced.json`	its sealed scores, all scopes and CIs
`probe/doc_feasibility.json`	frontier walk for both gates
`probe/gate_augment.json`	the synthetic-negative result
`decision/headline_balanced.json`	policy applied to arm B and the balanced gate
`training/h2h_cascade_rebuild.py` · `h2h_feasibility.py`	the two new modules

Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
`fusion-1k`	10360_betterdataai_ner_silver_eval_10.36k				0.208	1360.8	0.7
`fusion-1k`	10626_ai4privacy_pii_masking_eval_10.63k				0.424	1360.8	0.7
`fusion-1k`	20000_pii_holdout_20.00k				0.477	1360.8	0.7
`fusion-1k`	30000_pii2_eval_25.15k				0.413	1360.8	0.7
`fusion-1k`	38937_openpii_pii_eval_38.94k				0.510	1360.8	0.7
`fusion-1k`	4000_datax-dualjudge-evalset-1.32k					1360.8	0.7

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
`fusion-1k`	5617_nemotron_eval_5.36k					1360.8	0.7
`fusion-1k`	6589_govdocs2-dualjudge-eval20-3.53k					1360.8	0.7
`steady-cascade`	10360_betterdataai_ner_silver_eval_10.36k				0.266	314.1	3.2
`steady-cascade`	10626_ai4privacy_pii_masking_eval_10.63k				0.729	314.1	3.2
`steady-cascade`	20000_pii_holdout_20.00k				0.709	314.1	3.2
`steady-cascade`	30000_pii2_eval_25.15k				0.757	314.1	3.2
`steady-cascade`	38937_openpii_pii_eval_38.94k				0.657	314.1	3.2
`steady-cascade`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`steady-cascade`	5617_nemotron_eval_5.36k					314.1	3.2
`steady-cascade`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`fusion-12k`	10360_betterdataai_ner_silver_eval_10.36k				0.208	270.3	3.7
`fusion-12k`	10626_ai4privacy_pii_masking_eval_10.63k				0.425	270.3	3.7
`fusion-12k`	20000_pii_holdout_20.00k				0.485	270.3	3.7
`fusion-12k`	30000_pii2_eval_25.15k				0.390	270.3	3.7
`fusion-12k`	38937_openpii_pii_eval_38.94k				0.511	270.3	3.7
`fusion-12k`	4000_datax-dualjudge-evalset-1.32k					270.3	3.7
`fusion-12k`	5617_nemotron_eval_5.36k					270.3	3.7
`fusion-12k`	6589_govdocs2-dualjudge-eval20-3.53k					270.3	3.7

Measured 2026-08-25 · n=10,360, n=10,626, n=20,000, n=30,000, n=38,937, n=4,000, n=5,617, n=6,589.